

Stock Analysis

By Anshika Goel, Applied Data Science Department | San Jose State University

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Overview

The stock app enables user to retrieve historical stock data via web scraping or by importing a csv file. A user can add/update/delete/list stock in console as well as GUI application. Functionalities like displaying historical data, report and trendline have also been included in order to effectively analyze stock data and enforce data driven decisions.

GitHub Link: https://github.com/GoelAnshika99/DATA200_Lab2

Let's go through the features of the application step by step.

Main Menu

A menu is displayed on the console for user to make choice if they want to continue with the console application or switch over to GUI application.

```
Welcome to the Stock Manager Program!  
Please choose an option:  
1 - Start Console Version  
2 - Start GUI Version  
0 - Exit Program  
Enter your choice: 
```

- **Console Application**

A sub menu is displayed once the console application is selected.

```
Stock Analyzer ---  
1 - Manage Stocks (Add, Update, Delete, List)  
2 - Add Daily Stock Data (Date, Price, Volume)  
3 - Show Report  
4 - Show Chart  
5 - Manage Data (Save, Load, Retrieve)  
0 - Exit Program  
Enter Menu Option: 
```

1. **Manage Stocks**

The option manage stocks allows a user to add/update/delete/list stocks.

```

Manage Stocks ---
1 - Add Stock
2 - Update Shares
3 - Delete Stock
4 - List Stocks
0 - Exit Manage Stocks
Enter Menu Option: █

```

a) Add Stock

This allows user to add stock details like how many symbols, name and the shares a user owns.

```

Manage Stocks ---
1 - Add Stock
2 - Update Shares
3 - Delete Stock
4 - List Stocks
0 - Exit Manage Stocks
Enter Menu Option: 1
Enter Stock Symbol: GOOG
Enter Stock Name: Google
Enter Number of Shares: 23
Stock GOOG added successfully!
Press Enter to continue.....█

```

```

Manage Stocks ---
1 - Add Stock
2 - Update Shares
3 - Delete Stock
4 - List Stocks
0 - Exit Manage Stocks
Enter Menu Option: 1
Enter Stock Symbol: MSFT
Enter Stock Name: Microsoft
Enter Number of Shares: 200
Stock MSFT added successfully!
Press Enter to continue.....█

```

b) Update Shares

A user has the ability to either buy or sell stocks for the symbol entered into the console application. Make sure to save the data to database to reflect the changes.

```

Manage Stocks ---
1 - Add Stock
2 - Update Shares
3 - Delete Stock
4 - List Stocks
0 - Exit Manage Stocks
Enter Menu Option: 2
Enter Stock Symbol to update: MSFT
1 - Buy Shares
2 - Sell Shares
Enter Option: 1
Enter the number of shares to buy: 100
100.0 shares added to MSFT. Total shares: 300.0
Press Enter to continue....█

```

```
3 select * from stocks
```

	symbol	name	shares
1	GOOG	Google	23.0
2	MSFT	Microsoft	300.0

Buy Stock

```

Manage Stocks ---
1 - Add Stock
2 - Update Shares
3 - Delete Stock
4 - List Stocks
0 - Exit Manage Stocks
Enter Menu Option: 2
Enter Stock Symbol to update: MSFT
1 - Buy Shares
2 - Sell Shares
Enter Option: 2
Enter number of shares to sell: 250
250.0 shares sold from MSFT. Total shares: 50.0
Press Enter to continue....█

```

```
3 select * from stocks
```

	symbol	name	shares
1	GOOG	Google	23.0
2	MSFT	Microsoft	50.0

Sell Stock

c) Delete Stock

If a user no longer wishes to maintain data for a particular symbol, then they can delete stock data. The stock data will be deleted from both memory as well as database.

```
Manage Stocks ---
1 - Add Stock
2 - Update Shares
3 - Delete Stock
4 - List Stocks
0 - Exit Manage Stocks
Enter Menu Option: 3
Enter Stock Symbol to delete: MSFT
Stock MSFT and its data deleted from list.
Stock MSFT and its data deleted from the database.
Press Enter to continue....
```

```
select * from stocks
```

	symbol	name	shares
1	GOOG	Google	23.0

d) List Stocks

User has the option to view what all stocks do they own at a glance and further make decisions based on the data represented.

```
Tracked Stocks:
Symbol: GOOG, Name: Google, Shares: 23.0
Symbol: MSFT, Name: Microsoft, Shares: 200.0
Press Enter to continue....
```

2. Add Daily Stock Data

Apart from importing data through CSV and web scraping, user can also enter stock data manually.

```
Stock Analyzer ---
1 - Manage Stocks (Add, Update, Delete, List)
2 - Add Daily Stock Data (Date, Price, Volume)
3 - Show Report
4 - Show Chart
5 - Manage Data (Save, Load, Retrieve)
0 - Exit Program
Enter Menu Option: 2
Enter Stock Symbol to add data to: MSFT
Enter Date (MM/DD/YYYY): 02/01/2024
Enter Price: 23
Enter Volume: 45
Data for MSFT on 02/01/2024 added successfully!
Press Enter to continue....
```

```
select * from dailyData where symbol = 'MSFT' order by date desc
```

	symbol	date	price	volume
1	MSFT	02/01/24	23.0	45.0
2	MSFT	01/30/24	404.75	33477600.0
3	MSFT	01/29/24	405.87	24510200.0
4	MSFT	01/26/24	400.13	17803300.0
5	MSFT	01/25/24	401.06	21021200.0
6	MSFT	01/24/24	398.77	24867000.0
7	MSFT	01/23/24	395.15	20525900.0
8	MSFT	01/22/24	392.78	27016900.0

Execution finished without errors.
Result: 21 rows returned in 4ms
At line 3:
select * from dailyData where symbol = 'MSFT' order by date desc

3. Show Report

This option allows user to view daily price and volume of stocks grouped by symbol.

```

Stock Report ---

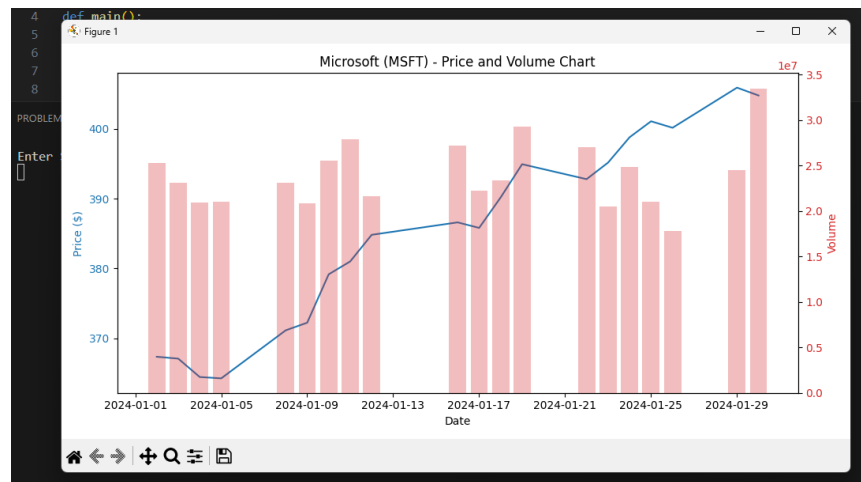
Stock: GOOG, Google
Date: 01/02/2024, Price: 138.9, Volume: 20071900.0
Date: 01/03/2024, Price: 139.7, Volume: 18974300.0
Date: 01/04/2024, Price: 137.39, Volume: 18253300.0
Date: 01/05/2024, Price: 136.74, Volume: 15433200.0
Date: 01/08/2024, Price: 139.87, Volume: 17645300.0
Date: 01/09/2024, Price: 141.89, Volume: 19579700.0
Date: 01/10/2024, Price: 143.12, Volume: 16641900.0
Date: 01/11/2024, Price: 142.99, Volume: 17471100.0
Date: 01/12/2024, Price: 143.56, Volume: 13995200.0
Date: 01/16/2024, Price: 143.4, Volume: 19198900.0
Date: 01/17/2024, Price: 142.22, Volume: 17884500.0
Date: 01/18/2024, Price: 144.31, Volume: 18876800.0
Date: 01/19/2024, Price: 147.27, Volume: 27170900.0
Date: 01/22/2024, Price: 147.01, Volume: 21829200.0
Date: 01/23/2024, Price: 147.98, Volume: 14113600.0
Date: 01/24/2024, Price: 149.64, Volume: 19245000.0
Date: 01/25/2024, Price: 152.92, Volume: 21495100.0
Date: 01/26/2024, Price: 153.07, Volume: 19494500.0
Date: 01/29/2024, Price: 154.11, Volume: 20909300.0
Date: 01/30/2024, Price: 152.33, Volume: 26578900.0

Stock: MSFT, Microsoft
Date: 01/02/2024, Price: 367.38, Volume: 25258600.0
Date: 01/03/2024, Price: 367.11, Volume: 23083500.0
Date: 01/04/2024, Price: 364.48, Volume: 20901500.0
Date: 01/05/2024, Price: 364.29, Volume: 20987000.0
Date: 01/08/2024, Price: 371.16, Volume: 23134000.0
Date: 01/09/2024, Price: 372.25, Volume: 20830000.0
Date: 01/10/2024, Price: 379.17, Volume: 25514200.0
Date: 01/11/2024, Price: 381.01, Volume: 27850800.0
Date: 01/12/2024, Price: 384.82, Volume: 21645700.0

```

4. Show Chart

A chart showcasing the daily trend of price and volumes is displayed.



5. Manage Data

The option manage data allows user to perform operations like save/load/retrieve/import.

```

1 - Manage Stocks (Add, Update, Delete, List)
2 - Add Daily Stock Data (Date, Price, Volume)
3 - Show Report
4 - Show Chart
5 - Manage Data (Save, Load, Retrieve)
0 - Exit Program
Enter Menu Option: 5
Manage Data ---
1 - Save Data to Database
2 - Load Data from Database
3 - Retrieve Stock Data from Web
4 - Import Data from CSV
0 - Exit Manage Data
Enter Menu Option: 

```

a) Save Data to Database

After completing the changes in memory, a user is required to explicitly save data to database otherwise changes made will not be reflected.

```
Manage Data ---
1 - Save Data to Database
2 - Load Data from Database
3 - Retrieve Stock Data from Web
4 - Import Data from CSV
0 - Exit Manage Data
Enter Menu Option: 1
Data saved to database.
```

b) Load Data from Database

Once the user exits application, the data is no longer available in the memory and therefore it cannot be modified or displayed. Before performing any operations user must ensure to load data from database to memory.

```
Manage Data ---
1 - Save Data to Database
2 - Load Data from Database
3 - Retrieve Stock Data from Web
4 - Import Data from CSV
0 - Exit Manage Data
Enter Menu Option: 2

Stocks loaded from database:
GOOG - Google - 23.0 shares
01/02/2024 - $138.90 - Volume: 20071900.0
01/03/2024 - $139.70 - Volume: 18974300.0
01/04/2024 - $137.39 - Volume: 18253300.0
01/05/2024 - $136.74 - Volume: 15433200.0
01/08/2024 - $139.87 - Volume: 17645300.0
01/09/2024 - $141.89 - Volume: 19579700.0
01/10/2024 - $143.12 - Volume: 16641900.0
01/11/2024 - $142.99 - Volume: 17471100.0
01/12/2024 - $143.56 - Volume: 13995200.0
01/16/2024 - $143.40 - Volume: 19198900.0
01/17/2024 - $142.22 - Volume: 17884500.0
01/18/2024 - $144.31 - Volume: 18876800.0
01/19/2024 - $147.27 - Volume: 27170900.0
01/22/2024 - $147.01 - Volume: 21829200.0
01/23/2024 - $147.98 - Volume: 14113600.0
01/24/2024 - $149.64 - Volume: 19245000.0
01/25/2024 - $152.92 - Volume: 21495100.0
01/26/2024 - $153.07 - Volume: 19494500.0
01/29/2024 - $154.11 - Volume: 20909300.0
01/30/2024 - $152.33 - Volume: 26578900.0
MSFT - Microsoft - 200.0 shares
01/02/2024 - $367.38 - Volume: 25258600.0
01/03/2024 - $367.11 - Volume: 23083500.0
01/04/2024 - $364.48 - Volume: 20901500.0
01/05/2024 - $364.29 - Volume: 20987000.0
01/08/2024 - $371.16 - Volume: 23134000.0
01/09/2024 - $372.25 - Volume: 20830000.0
01/10/2024 - $379.17 - Volume: 25514200.0
```

c) Retrieve Stock Data from Web

This option enables user to retrieve stock data from yahoo finance website. User is required to input the date range and it will return the number of rows fetched in that particular date range.

```

Manage Data ---
1 - Save Data to Database
2 - Load Data from Database
3 - Retrieve Stock Data from Web
4 - Import Data from CSV
0 - Exit Manage Data
Enter Menu Option: 3
Enter start date (MM/DD/YYYY): 01/01/2024
Enter end date (MM/DD/YYYY): 01/31/2024
40 records retrieved from the web.

```

Data retrieved from web

```
5 select * from stocks
```

	symbol	name	shares
1	GOOG	Google	23.0
2	MSFT	Microsoft	200.0

```
3 select * from dailyData order by symbol, date
```

	symbol	date	price	volume
17	GOOG	01/25/24	152.92	21495100.0
18	GOOG	01/26/24	153.07	19494500.0
19	GOOG	01/29/24	154.11	20909300.0
20	GOOG	01/30/24	152.33	26578900.0
21	MSFT	01/02/24	367.38	25258600.0
22	MSFT	01/03/24	367.11	23083500.0
23	MSFT	01/04/24	364.48	20901500.0
24	MSFT	01/05/24	364.29	20987000.0

Execution finished without errors.
Result: 40 rows returned in 6ms
At line 3:
select * from dailyData order by symbol, date

Data from web saved to database

d) Import Data from CSV

Another option to fetch data into the stock application is through csv file. User is required to input the stock symbol and the file path. Please ensure the structure of the csv file is same as the image given below.

```

Stock Analyzer ---
1 - Manage Stocks (Add, Update, Delete, List)
2 - Add Daily Stock Data (Date, Price, Volume)
3 - Show Report
4 - Show Chart
5 - Manage Data (Save, Load, Retrieve)
0 - Exit Program
Enter Menu Option: 5
Manage Data ---
1 - Save Data to Database
2 - Load Data from Database
3 - Retrieve Stock Data from Web
4 - Import Data from CSV
0 - Exit Manage Data
Enter Menu Option: 4
Enter Stock Symbol to import CSV for: MSFT
Enter CSV filename: "C:\Users\aradh\Desktop\DATA 200 Python Programming\Lab2\stock.csv"
Data imported from C:/Users/aradh/Desktop/DATA 200 Python Programming/Lab2/stock.csv.
Manage Data ---
1 - Save Data to Database
2 - Load Data from Database
3 - Retrieve Stock Data from Web
4 - Import Data from CSV
0 - Exit Manage Data
Enter Menu Option: 

```

Data imported via csv file

symbol	date	price	volume
MSFT	4/29/2025	284.54	14978000
MSFT	4/28/2025	291.16	16774000
MSFT	4/25/2025	291.85	18973000
MSFT	4/24/2025	287.3	22112100
MSFT	4/23/2025	274.39	20545100
MSFT	4/22/2025	266.82	19480000
MSFT	4/21/2025	259.12	20807300
MSFT	4/17/2025	267.78	20941700
MSFT	4/16/2025	271.61	21967800
MSFT	4/15/2025	265.79	17799900
MSFT	4/14/2025	267.81	19251200
MSFT	4/13/2025	288.45	23832200
MSFT	4/10/2025	282.35	38534800
MSFT	4/9/2025	290.49	50195700
MSFT	4/8/2025	254.36	10880800
MSFT	4/7/2025	257.86	56620000
MSFT	4/6/2025	259.84	45209900
MSFT	4/2/2025	275.11	30180000
MSFT	4/1/2025	262.14	16952600
MSFT	4/1/2025	262.19	19481500
MSFT	3/31/2025	276.39	31148700
MSFT	3/28/2025	276.8	21812000
MSFT	3/27/2025	290.56	13756800
MSFT	3/26/2025	289.97	16108400
MSFT	3/25/2025	295.16	13770800
MSFT	3/24/2025	293.68	21050100
MSFT	3/23/2025	291.26	39870900
MSFT	3/20/2025	286.84	18670100
MSFT	3/19/2025	287.82	19185500
MSFT	3/18/2025	283.52	19480000
MSFT	3/17/2025	286.7	24593000
MSFT	3/14/2025	288.56	19913800
MSFT	3/13/2025	276.77	24672000
MSFT	3/12/2025	283.27	24253600
MSFT	3/11/2025	280.45	30380200
MSFT	3/10/2025	280.18	12840100
MSFT	3/7/2025	293.11	22034100

```
3 select * from dailyData
```

	symbol	date	price	volume
1	MSFT	04/29/25	72.35	16244100.0
2	MSFT	04/28/25	71.79	15997500.0
3	MSFT	04/25/25	71.91	16316000.0
4	MSFT	04/24/25	72.52	16893900.0
5	MSFT	04/23/25	73.3	16353900.0
6	MSFT	04/22/25	73.9	15613500.0
7	MSFT	04/21/25	72.77	16037200.0
8	MSFT	04/17/25	73.0	20746800.0

Execution finished without errors.
Result: 664 rows returned in 7ms
At line 1:
select * from dailyData

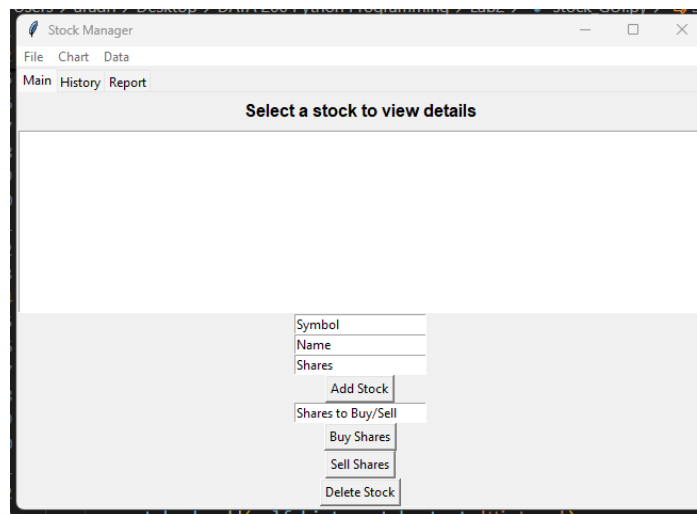
Data imported from csv and saved to database

1	symbol,date,price,volume
2	AMZN,04/29/25,187.39,41667300.0
3	AMZN,04/28/25,187.7,33224700.0
4	AMZN,04/25/25,188.99,36414300.0
5	AMZN,04/24/25,186.54,43763200.0
6	AMZN,04/23/25,180.6,63470100.0
7	AMZN,04/22/25,173.18,56607200.0
8	AMZN,04/21/25,167.32,48126100.0
9	AMZN,04/17/25,172.61,44468400.0
10	AMZN,04/16/25,174.33,51875300.0
11	AMZN,04/15/25,179.59,43642000.0
12	AMZN,04/14/25,182.12,48002500.0
13	AMZN,04/11/25,184.87,50594300.0
14	AMZN,04/10/25,181.22,68302000.0
15	AMZN,04/09/25,181.1,116004200.0

CSV file structure

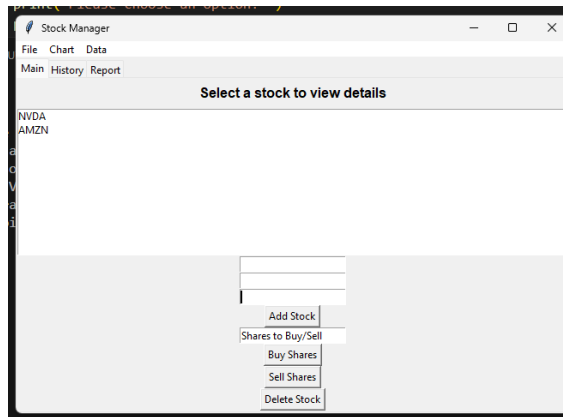
- **GUI Application**

The GUI stock application has various tabs and menu to perform the same operations as console stock application.



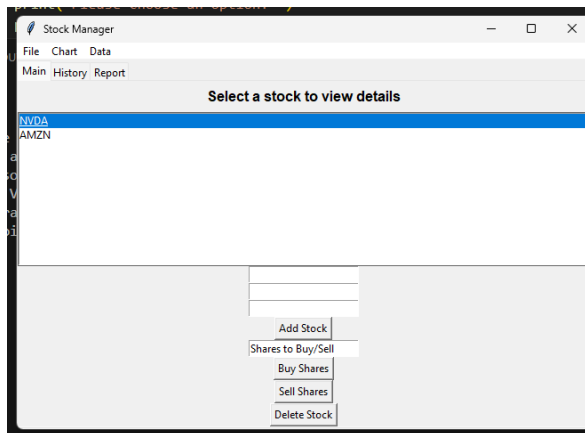
1. Main

The main tab lists the stocks that are available and also provide options to add/buy/sell/delete stocks.

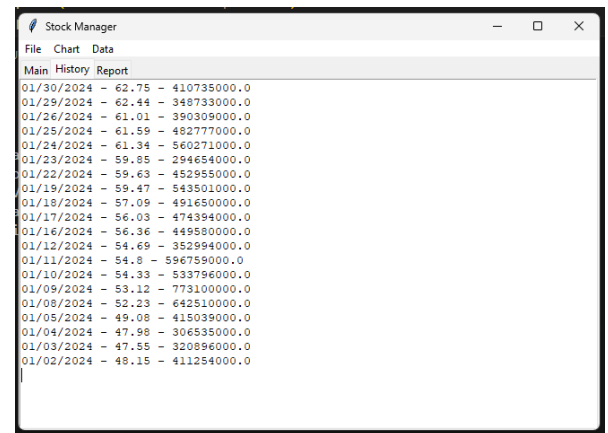


2. History

History tab displays the historical data if a stock is selected in the main tab.



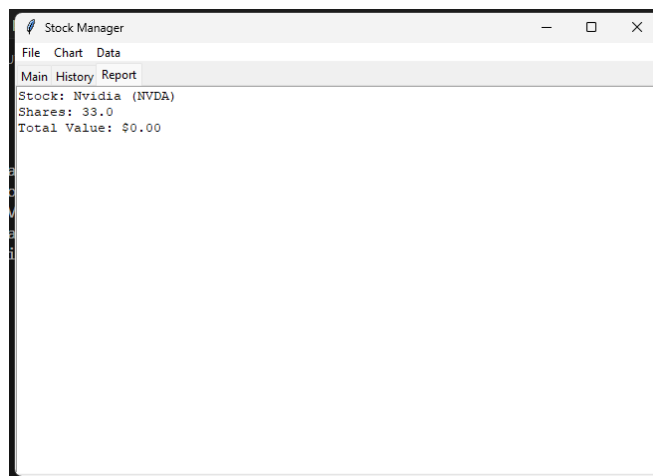
Make sure to select the stock in main tab



History of NVDA stock

3. Report

On selecting a stock in main tab, the report tab displays the current shares for a particular stock symbol

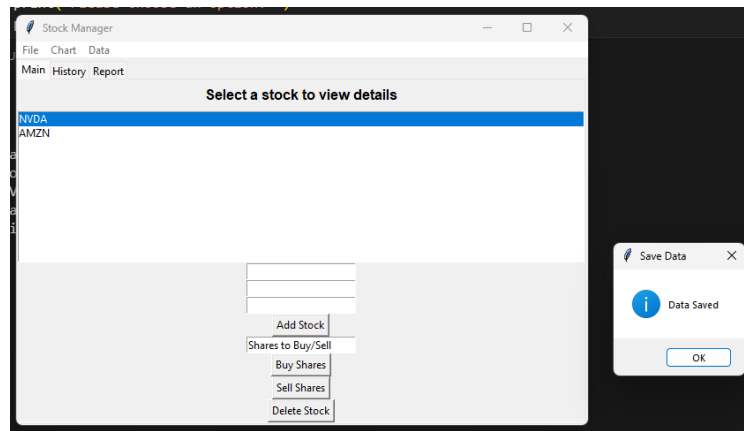


4. File

The file option in the menu bar enables user to perform the following main functions in the stock application:

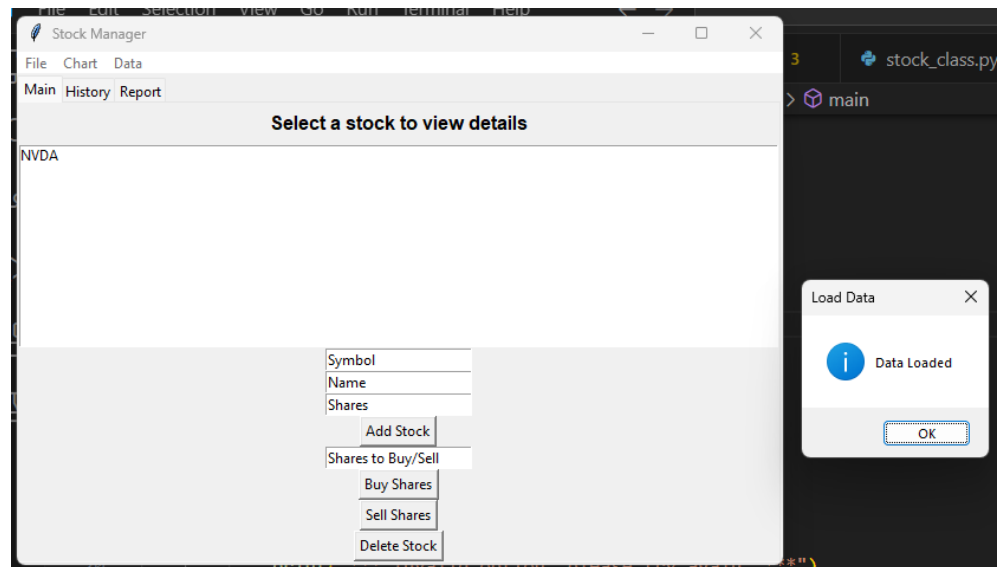
a) Save Data

Any changes made using GUI application are performed in memory and will not be reflected in the database automatically. This allows a user to explicitly save the changes to database.



b) Load Data

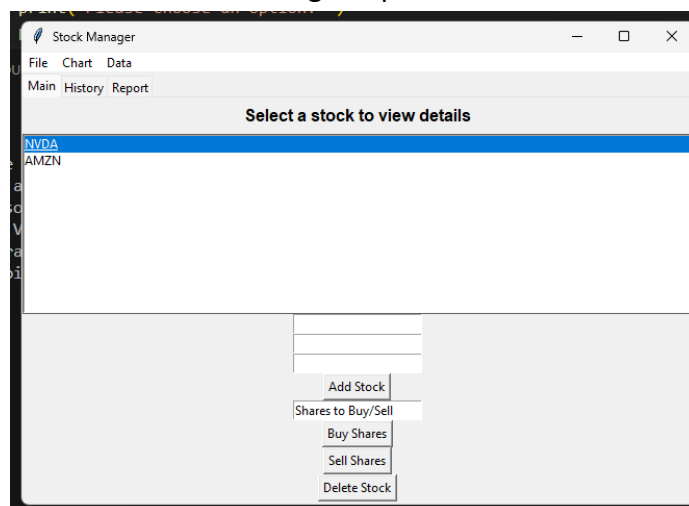
Once the GUI application is started user must ensure to load data from database. The GUI application utilizes memory to modify/display stock data therefore, data must be loaded from database into memory before performing any operations.



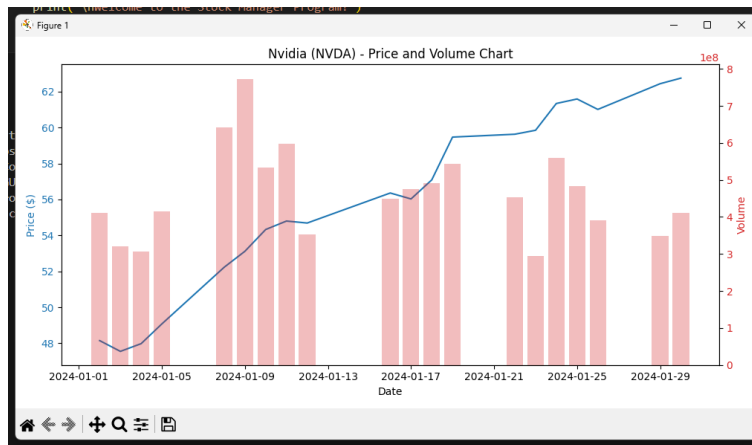
5. Chart

a) Show Chart

A stock should be selected before viewing the price and volume trend over date.



Make sure to select the stock in main tab



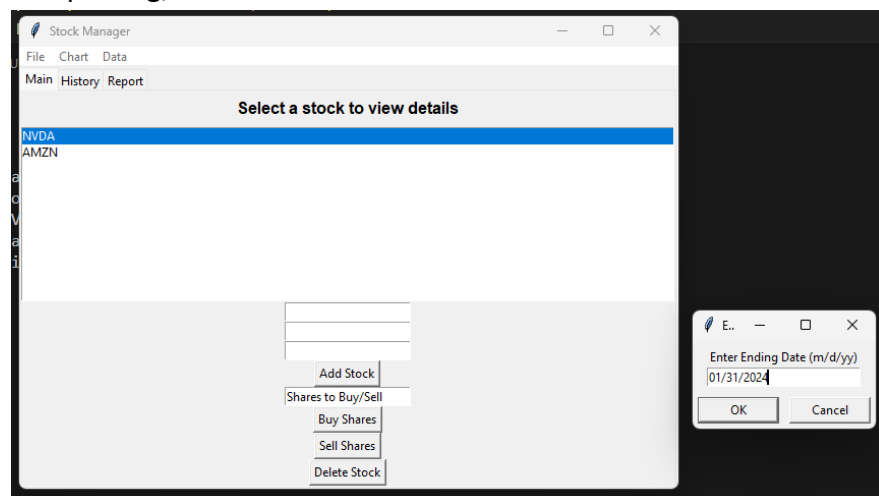
Price and Volume Chart for NVDA

6. Data

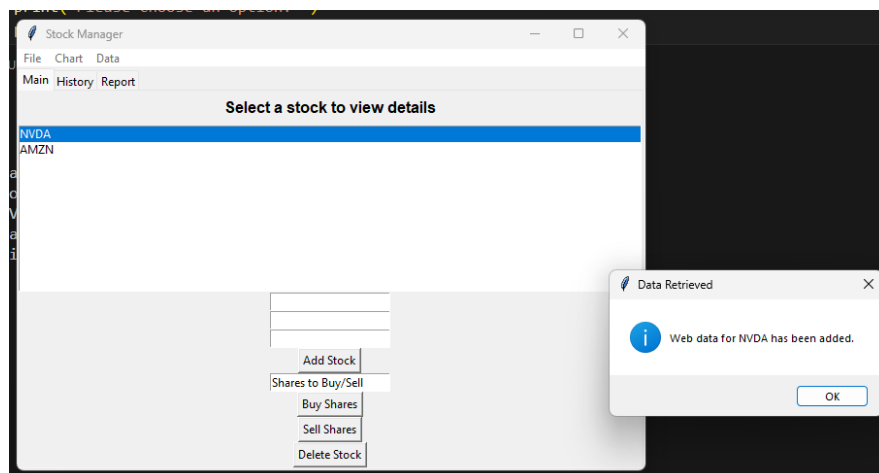
The data option in menu bar provides functionality to add daily data either by web scraping or by importing a csv

a) Import from Web

This option enables user to import data from web for a particular symbol and date range. After importing, make sure to save data into database.



Input date range for symbol NVDA



Web data successfully added for symbol NVDA

3	select * from stocks		
	symbol	name	shares
1	NVDA	Nvidia	33.0
2	AMZN	Amazon	100.0

3 select * from dailyData

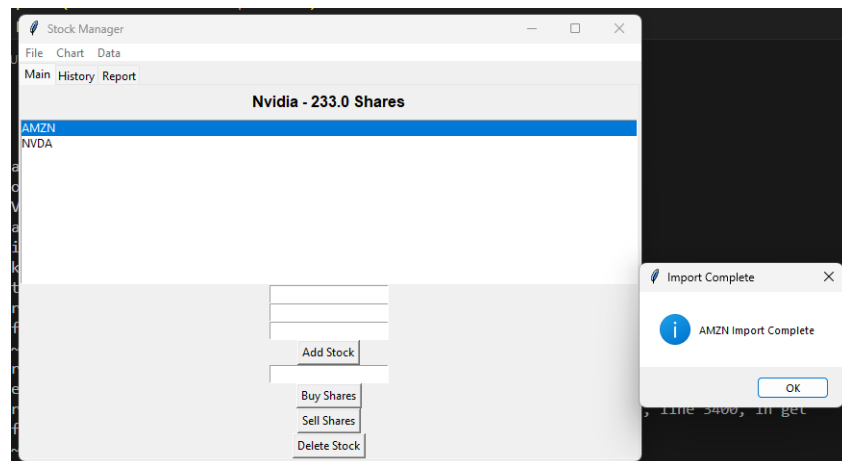
	symbol	date	price	volume
17	NVDA	01/05/24	49.08	415039000.0
18	NVDA	01/04/24	47.98	306535000.0
19	NVDA	01/03/24	47.55	320896000.0
20	NVDA	01/02/24	48.15	411254000.0
21	AMZN	01/30/24	159.0	45207400.0
22	AMZN	01/29/24	161.26	45270400.0
23	AMZN	01/26/24	159.12	51047400.0
24	AMZN	01/25/24	157.75	43638600.0

Execution finished without errors.
Result: 40 rows returned in 4ms
At line 3:
select * from dailyData

Data saved in database for symbol NVDA and AMZN using web scraping

b) Import from csv

This option enables user to import data through csv file.



Daily data for symbol AMZN and NVDA imported via csv file

1	Symbol	Date	Price	Volume
117	AMZN	1/25/2024	157.75	43638600
118	AMZN	1/24/2024	156.87	48547300
119	AMZN	1/23/2024	156.02	37986000
120	AMZN	1/22/2024	154.78	43687500
121	AMZN	1/19/2024	155.34	51033700
122	AMZN	1/18/2024	153.5	37802000
123	AMZN	1/17/2024	151.71	34953400
124	AMZN	1/16/2024	153.16	41384600
125	AMZN	1/12/2024	154.62	40480300
126	AMZN	1/11/2024	155.18	49072700
127	AMZN	1/10/2024	153.79	44421800
128	AMZN	1/9/2024	153.37	43812600
129	AMZN	1/8/2024	149.1	46757100
130	AMZN	1/5/2024	149.04	45124800
131	AMZN	1/4/2024	144.37	56078000
132	AMZN	1/3/2024	148.47	49425500
133	AMZN	1/2/2024	149.93	47139400
1944	NVDA	4/29/2025	209.92	17044300
1945	NVDA	4/28/2025	198.73	20778500
1946	NVDA	4/25/2025	111.01	251084700
1947	NVDA	4/24/2025	106.43	220815000
1948	NVDA	4/23/2025	102.71	247520000
1949	NVDA	4/22/2025	98.89	241004800
2000	NVDA	4/21/2025	96.91	288501100
2001	NVDA	4/17/2025	101.49	290333400
2002	NVDA	4/16/2025	104.49	397618900
2003	NVDA	4/15/2025	112.2	228966900
2004	NVDA	4/14/2025	110.71	284705000
2005	NVDA	4/11/2025	110.93	313417500
2006	NVDA	4/10/2025	107.37	417813400
2007	NVDA	4/9/2025	114.33	612918300
2008	NVDA	4/8/2025	96.3	476343400
2009	NVDA	4/7/2025	97.64	411941300
2010	NVDA	4/6/2025	94.31	532273800
2011	NVDA	4/3/2025	101.8	138769400
2012	NVDA	4/2/2025	110.42	22601200
2013	NVDA	4/1/2025	110.15	22904400

Rows 664 available for symbol NVDA and AMZN in the csv file

```

3 select * from dailyData

```

	symbol	date	price	volume
1	AMZN	04/29/25	72.35	16244100.0
2	AMZN	04/28/25	71.79	15997500.0
3	AMZN	04/25/25	71.91	16316000.0
4	AMZN	04/24/25	72.52	16893900.0
5	AMZN	04/23/25	73.3	16353900.0
6	AMZN	04/22/25	73.9	15613500.0
7	AMZN	04/21/25	72.77	16037200.0
8	AMZN	04/17/25	73.0	20746800.0

```

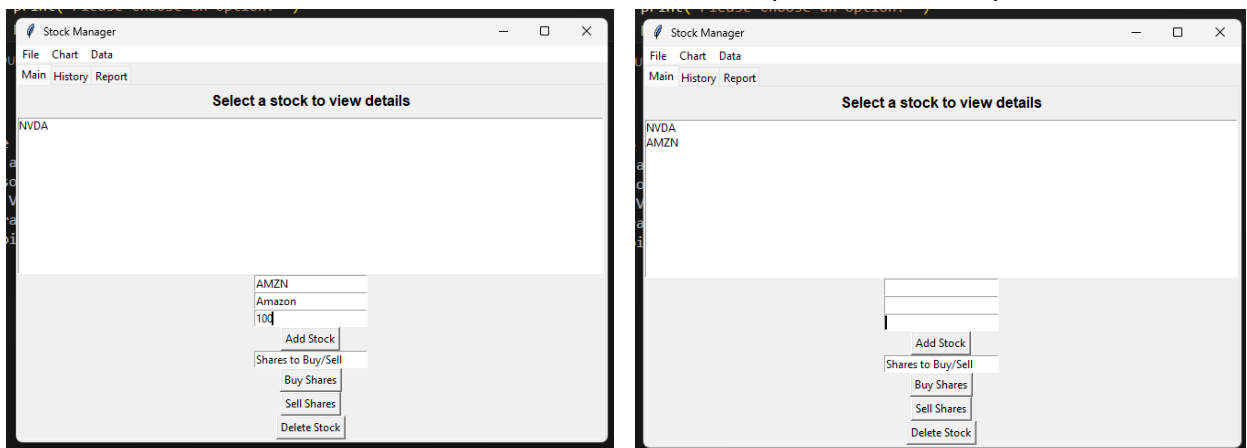
Execution finished without errors.
Result: 664 rows returned in 6ms
At line 3:
select * from dailyData

```

664 rows saved to dailyData for symbol NVDA and AMZN in database

7. Add Stock

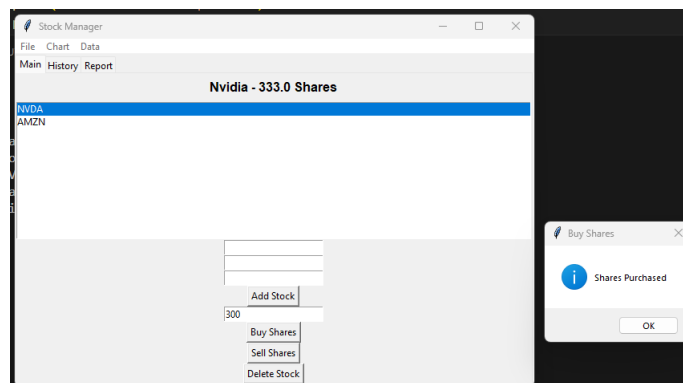
User can add stocks and the number of shares owned for a particular stock symbol.



Stock symbol AMZN added to stock application

8. Buy Share

This option enables user to buy additional shares and can save the information in database using save data option in file menu.



```

3 select * from stocks

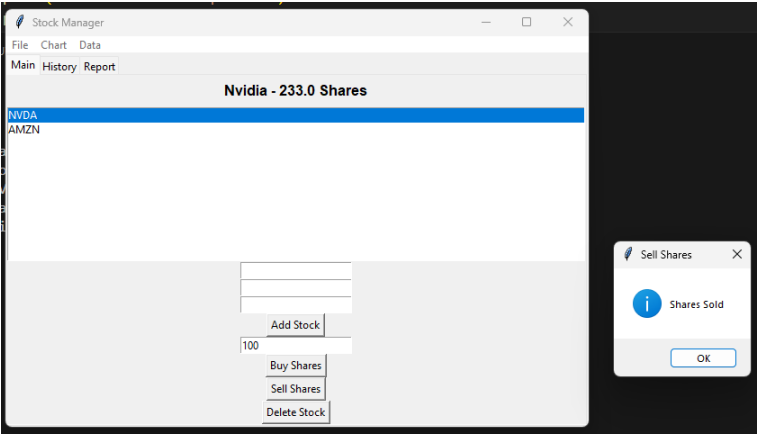
```

	symbol	name	shares
1	NVDA	Nvidia	333.0
2	AMZN	Amazon	100.0

Shares purchased for symbol NVDA and the same is reflected in database

9. Sell Shares

This option enables user to sell shares and can save the information in database using save data option in file menu.



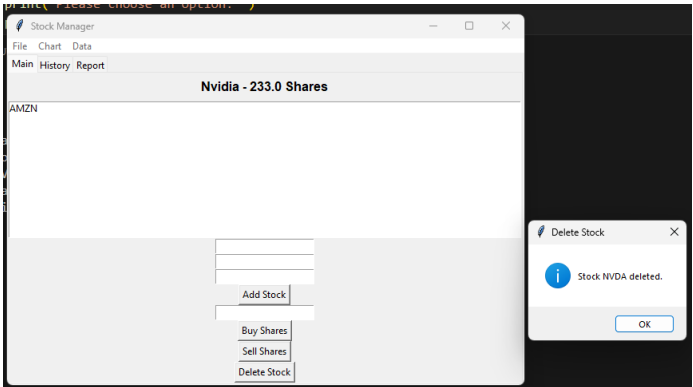
```
3 select * from stocks
```

	symbol	name	shares
1	NVDA	Nvidia	233.0
2	AMZN	Amazon	100.0

Shares sold for symbol NVDA and the same is reflected in database

10. Delete Stock

Allows user to delete the stock symbol if no longer required.



```
3 select * from stocks
```

	symbol	name	shares
1	AMZN	Amazon	100.0

Stock symbol NVDA deleted